

WEEK 001

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Let $\mathcal{C}$ be the collection of all finite subsets of $\mathbb{N}$, and let $\mathcal{C}_n$ be the set of all subsets of $\mathbb{N}$ of n elements, where $n \in \mathbb{N}$.

Theorem 1 (Schröder-Bernstein). *Suppose $f: A \rightarrow B$ and $g: B \rightarrow A$ are injections. Then there exists a bijection from A to B .*

(a) *Show that $|\mathcal{C}_k| = |\mathbb{N}^k|$ for all positive integers k .*

Consider the function $ord: \mathcal{C}_k \rightarrow \mathbb{N}^k$ which takes an element $X \in \mathcal{C}_k$ to the k -tuple consisting of the elements of X in order. This is an injection since we can invert it: take the set containing each component of the k -tuple.

Consider the function $\mathbb{N}^k \rightarrow \mathcal{C}_k$ given by the map

$$(n_1, n_2, \dots, n_k) \mapsto \{(r-1) + \sum_{i=1}^r n_i | r \in \{1, \dots, k\}\}.$$

Note the r -th term strictly exceeds the $(r-1)$ -th term, since their difference is $1 + n_r \geq 1$; hence the set has exactly k elements and the map indeed lands in $\mathcal{C}_k$.

This function is an injection since we can invert it:

$$\begin{aligned} \{(r-1) + \sum_{i=1}^r n_i | r \in \{1, \dots, k\}\} &\xrightarrow{ord} (n_1, 1 + n_1 + n_2, \dots, k-1 + \sum_{i=1}^k n_i) \\ &\xrightarrow{dif} (n_1, n_2 + 1, \dots, n_k + 1) \\ &\xrightarrow{red} (n_1, n_2, \dots, n_k) \end{aligned}$$

where the *dif* map replaces the later terms with their difference to their previous term and *red* reduces the result by subtracting 1 from every term except the first.

Then $|\mathcal{C}_k| = |\mathbb{N}^k|$ by applying Schröder-Bernstein with these injections.

(b) *Show, explicitly, that there is a bijection between $\mathbb{N}^2$ and $\mathbb{N}$.*

Consider the function $diag: \mathbb{N}^2 \rightarrow \mathbb{N}$ defined as follows:

$$(m, n) \mapsto n + \sum_{i=1}^{m+n} i.$$

One can think of this function as ordering $\mathbb{N}^2$ primarily by the sum of components, with a fall back to the second component (see figure 1).

] As such, we can easily construct an inverse map $diag^{-1}: \mathbb{N} \rightarrow \mathbb{N}^2$. Indeed, to find $(m, n) := diag^{-1}(x)$ we first let w be the index of the largest triangle number

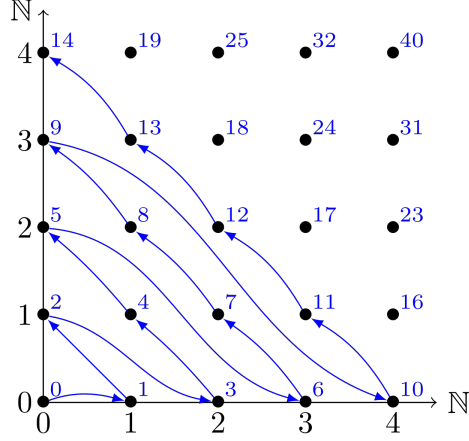


FIGURE 1. A diagram showing the ordering perspective.

at most x . Then let t be the w th triangle number. Then we know that $n = x - t$ and $m = w - n$. Putting these operations together forms a simple function which is easy to observe is the inverse of $diag$.

Unlike the previous inverses we've constructed, this acts on any element $n \in \mathbb{N}$, making it a total inverse rather than a partial inverse, and hence proving both functions are bijections, and thus that $|\mathbb{N}^2| = |\mathbb{N}|$.

(c) Show that $|\mathbb{N}^k| = |\mathbb{N}|$ for all positive integers k by induction.

To prove this by induction, the base case $k = 1$ is trivial. For the inductive step, we assume $|\mathbb{N}^k| = |\mathbb{N}|$.

That is, we have a bijection $\mathbb{N}^k \xleftrightarrow{f_k} \mathbb{N}$. Then we can compose bijections as follows, to obtain a bijection $\mathbb{N}^{k+1} \xleftrightarrow{f_{k+1}} \mathbb{N}$:

$$\mathbb{N}^{k+1} = \mathbb{N} \times \mathbb{N}^k \xleftrightarrow{\text{id} \times f_k} \mathbb{N} \times \mathbb{N} = \mathbb{N}^2 \xleftrightarrow{diag} \mathbb{N}.$$

(d) By considering the fact that $\mathcal{C} = \bigcup_{n \in \mathbb{N}} \mathcal{C}_n$, show that $|\mathcal{C}| = |\mathbb{N}|$.

First, we observe that $|\mathcal{C}_n| = |\mathbb{N}|$ for each natural n (except 0) by composing the equalities from the previous parts. Thus, by observing that the union $\bigcup_{n \in \mathbb{N}} \mathcal{C}_n$ is a disjoint union, we may immediately get

$$|\mathcal{C}| = \left| \{\emptyset\} \sqcup \left(\bigsqcup_{n \in \mathbb{N}^+} \mathbb{N} \right) \right|.$$

Formally,

$$\bigsqcup_{n \in \mathbb{N}^+} \mathbb{N} := \{(n, x) \mid n \in \mathbb{N}^+, x \in \mathbb{N}\} = \mathbb{N}^+ \times \mathbb{N}$$

Since $|\mathbb{N}^+| = |\mathbb{N}|$, $|\mathbb{N}^+ \times \mathbb{N}| = |\mathbb{N}^2| = |\mathbb{N}|$. So, $|\mathcal{C}| = |\{\emptyset\} \sqcup \mathbb{N}|$. Thus, $|\mathcal{C}| = |\mathbb{N}|$.

Q.E.D.